
Session 3 — Linear Algebra I

Exercise sheet

Work through these in order: they start with simple vector arithmetic and build toward systems, inverses, and rank. Keep the geometry in mind — vectors are arrows and points, and a matrix is both a table of data and a transformation of space. Show your calculations.

Exercise 1 Let $\mathbf{u} = (2, -1)$ and $\mathbf{v} = (1, 3)$. Compute:

- (a) $3\mathbf{u} - 2\mathbf{v}$;
 - (b) the length $\|\mathbf{u}\|$;
 - (c) the unit vector in the direction of $\mathbf{u}$.
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Exercise 2 Express the vector $(5, 1)$ as a linear combination of $(1, 1)$ and $(1, -1)$; that is, find a and b with $a(1, 1) + b(1, -1) = (5, 1)$. What does it mean, geometrically, that this is always possible for any target vector in the plane?

Exercise 3 *Two stimuli, each encoded by the activity of three neurons (a population vector).*

Let $\mathbf{a} = (3, 4, 0)$ and $\mathbf{b} = (0, 4, 3)$.

- (a) Compute the dot product $\mathbf{a} \cdot \mathbf{b}$ and the two lengths.
 - (b) Compute the cosine similarity $\cos \theta$, and the approximate angle.
 - (c) In one sentence, what does a high cosine similarity say about the two neural representations?
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Exercise 4 Two stimuli are represented by the vectors $\mathbf{p} = (1, 2, 2)$ and $\mathbf{q} = (3, 2, 4)$. Compute the Euclidean distance $\|\mathbf{q} - \mathbf{p}\|$ between their representations.

Exercise 5 Let $\mathbf{a} = (2, 1, -1)$ and $\mathbf{b} = (1, -1, 1)$.

- (a) Compute $\mathbf{a} \cdot \mathbf{b}$. Are the two vectors orthogonal?
 - (b) If two population vectors are orthogonal, what does that say about how the two stimuli are encoded?
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Exercise 6 *Projection is the geometric heart of least-squares regression.*

Project $\mathbf{b} = (4, 2)$ onto $\mathbf{a} = (3, 0)$, using $\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}$.

- (a) Compute the projection.
 - (b) Compute the residual $\mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b}$ and verify it is orthogonal to $\mathbf{a}$.
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Exercise 7 *A linear readout maps a 2-feature input to a 2-neuron response via a weight matrix.*

Let $\mathbf{W} = \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix}$ and the input $\mathbf{x} = (1, 2)$. Compute the response $\mathbf{W}\mathbf{x}$.

Exercise 8 Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ (a shear) and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ (a scaling).

- (a) Compute $\mathbf{AB}$.
 - (b) Compute $\mathbf{BA}$.
 - (c) Are they equal? What does this illustrate about composing transformations?
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Exercise 9 Let $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 2 & 4 \end{pmatrix}$.

- (a) Compute $\det \mathbf{A}$.
 - (b) By what factor does $\mathbf{A}$ scale areas?
 - (c) Is $\mathbf{A}$ invertible? How do you know?
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Exercise 10 Compute the inverse of $\mathbf{A} = \begin{pmatrix} 4 & 3 \\ 5 & 4 \end{pmatrix}$, and verify your answer by checking that $\mathbf{AA}^{-1} = \mathbf{I}$.

Exercise 11 *A simplified excitation–inhibition balance gives two linear equations.*

Solve the system $3x + 2y = 12$, $x + 2y = 8$.

Exercise 12 Solve the 3×3 system by elimination:

$$x + y + z = 6, \quad 2x - y + z = 3, \quad x + 2y - z = 2.$$

Exercise 13 Consider the three vectors $(1, 0, 0)$, $(0, 1, 0)$, and $(1, 1, 0)$.

- (a) Are they linearly independent? Justify.
- (b) What is their rank, and what does that say about the space they span?

Exercise 14 *Dimensionality of a small neural population.*

Across many trials, the activity vectors of three neurons always satisfy exactly $\mathbf{n}_3 = 2\mathbf{n}_1 - \mathbf{n}_2$.

- (a) Are the three neurons' activities linearly independent?
 - (b) What is the effective dimensionality (rank) of the population activity?
 - (c) What does this redundancy suggest about the neural code?
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Exercise 15 Let $\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

- (a) Compute $\mathbf{A}(1, 0)$ and $\mathbf{A}(0, 1)$. What transformation does $\mathbf{A}$ perform?
- (b) Compute $\mathbf{A}^2$ and interpret it geometrically.